

Abhijeet Palanki

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TARGET JOB TITLES

Frontend Developer | Senior Frontend Engineer | React Developer | React Native Developer
| Mobile Engineer | Software Engineer – Frontend

PROFESSIONAL SUMMARY

Senior Frontend & Mobile Engineer with 10+ years of experience building high-performance web and cross-platform applications using React.js, React Native, and TypeScript. Specialized in micro-frontend architecture, performance optimization, and scalable UI systems. Proven track record improving rendering speed for 1M+ users, reducing load times by 30%, and leading frontend initiatives across enterprise teams. Skilled in cloud deployment, CI/CD automation, and event-driven & AI-powered application workflows.

CORE SKILLS

- **Frontend:** React.js (Hooks, Suspense, Concurrent Features), React Native (iOS/Android), TypeScript, JavaScript (ES6+), Redux Toolkit, RTK Query, HTML5, CSS3, SCSS, Responsive UI
- **Mobile:** React Native CLI, Expo, Native Modules, iOS/Android Deployment
- **Architecture:** Micro-frontend Architecture, Runtime Dependency Injection, Event-Driven Systems, CQRS
- **Backend Integration:** Node.js, GraphQL, REST APIs
- **Testing:** Jest, React Testing Library, Playwright, Mocha, Automated Testing (Unit, Integration, E2E)
- **DevOps & Tooling:** Git, GitHub Actions, Docker, Kubernetes, CI/CD Pipelines, Webpack, Vite, Babel
- **Workflow & Design:** Agile/Scrum, JIRA, Figma, Storybook, Chromatic, OpenTelemetry

AI-DRIVEN FRONT-END & INTELLIGENT UI

- Specialized in building intelligent, AI-augmented UI systems that leverage embeddings, vector search, and real-time inference.
- Designed AI-powered UI workflows using OpenAI APIs, embeddings, and vector databases (Pinecone)

- Built semantic search, personalization engines, and AI-assisted catalog management tools.
- Integrated low-latency inference pipelines with frontend observability (OpenTelemetry)
- Developed context-aware UI components driven by embedding similarity and real-time signals.

PROFESSIONAL EXPERIENCE

Lead Software Developer | Sam's Club / Walmart, AR (July 2024 – Present)

- Developed and optimized high-performance React applications for internal productivity tools, improving rendering speed and data throughput for 1M+ daily active users.
- Architected scalable Redux state management with memoization, virtualization, and concurrency features, reducing unnecessary re-renders by 40%+.
- Improved application performance using React Suspense, batching, and concurrent rendering strategies, resulting in smoother UI interactions under heavy data loads.
- Ensured 85%+ automated test coverage using Jest, React Testing Library, and E2E frameworks, raising overall code quality and reducing regression defects.
- Collaborated with architects and backend teams to define scalable UI patterns, reusable components, and performance-first design principles.

Senior Software Developer | Tyler Technologies, NY (Dec 2020 – Apr 2024)

- Engineered enterprise-grade React applications using micro-frontend architecture, dynamic Module Federation, and runtime dependency injection for scalable deployments.
- Built automated testing pipelines with Jest, Mocha, and React Testing Library, enforcing strict coverage thresholds in CI/CD workflows.
- Containerized frontend workloads with Docker and deployed to Kubernetes clusters, enabling horizontal scaling and reducing deployment time by 30%.
- Partnered with product, UX, and backend teams to deliver complex features across multiple agile sprints, consistently meeting acceptance criteria and release deadlines.

Software Developer | State of Arkansas, AR (Aug 2018 – Dec 2020)

- Developed multi-screen React SPA applications using React Router v6 and TypeScript-based component libraries.
- Implemented automated unit and integration tests with Jest and Enzyme, reducing manual QA cycles by 25%.
- Built Node.js modules and REST APIs to integrate frontend workflows with backend services.
- Added PWA capabilities including service workers, offline caching, and push notifications to enhance reliability.
- Performed performance audits using Lighthouse and React Profiler, reducing Time to Interactive (TTI) by 30%.

Programmer Analyst | Walmart Stores Inc., AR (Jan 2016 – Jun 2018)

- Developed Angular 4 + TypeScript applications using reactive state patterns with RxJS.
- Applied AOT compilation, tree-shaking, and lazy loading to reduce bundle sizes and improve runtime performance.
- Automated testing with Jasmine and Karma, enforcing coverage gates in enterprise CI pipelines.
- Built Angular Material templates and implemented authentication flows, improving usability and reducing onboarding friction.

PROJECTS

- **Shirty - 3D T-Shirt Customizer** (A GPU-accelerated 3D apparel customization platform enabling real-time model rendering and texture manipulation)
 - Architected a WebGL pipeline using Three.js, custom GLSL shaders, and quaternion-based camera systems for smooth 3D interactions.
 - Optimized rendering performance to maintain 60fps on mid-range devices through shader tuning and mesh simplification.

Tech: Three.js, WebGL, GLSL, React, TypeScript

- **LiveDocs - Collaborative Document Editor** (A real-time document editing platform supporting multi-user collaboration with conflict-free syncing)
 - Built a CRDT-powered collaboration engine with WebSocket synchronization and SSR/ISR rendering for low-latency updates.
 - Integrated Sentry and OpenTelemetry for observability, reducing debugging time by 40%.

Tech: CRDTs, WebSockets, Next.js, Node.js, Sentry, OpenTelemetry

- **Whiz - Video Conferencing App** (A scalable video conferencing solution supporting adaptive streaming and real-time communication)
 - Engineered WebRTC media pipelines with adaptive bitrate algorithms, TURN/STUN signaling, and Next.js API routes for scalable streaming.
 - Implemented Next.js API routes for signaling and session management, reducing connection failures by 25%.

Tech: WebRTC, TURN/STUN, Next.js, React, Node.js

- **Gen-AI Item Catalog Maintenance** (AI-driven catalog enrichment and maintenance system for large-scale product datasets)
 - Designed vector-based search workflows using OpenAI embeddings and Pinecone for semantic retrieval.
 - Built low-latency inference pipelines with OpenTelemetry instrumentation for performance monitoring.

Tech: OpenAI, Pinecone, Node.js, Vector Search, OpenTelemetry

- **AI Personalization Engine** (2048 Animated Web Game - A fully animated, browser-based version of the classic 2048 puzzle game)

- Built smooth tile animations, transition effects, and responsive gameplay using React and TypeScript.
- Implemented optimized rendering logic, grid state management, and animation sequencing for fluid motion across devices.

Tech: React, TypeScript, JavaScript (ES6+), CSS Animations, HTML5 Canvas

EDUCATION

- Master of Science (M.S.) – Computer Science
[University of Bridgeport]
- Bachelor of Technology (B. Tech) – Electronics & Communication
[SRM University]